

Ensemble Learning

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What is Ensemble Learning?

Ensemble Learning is a machine learning technique that combines the predictions of multiple learning models (called **base learners**) to obtain better performance than a single model.

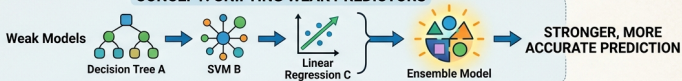
- Applicable to both classification and regression problems.
- Improves prediction accuracy.
- Reduces overfitting.
- Produces more robust and stable models.
- Inspired by the idea of “wisdom of the crowd.”

Ensemble Methods

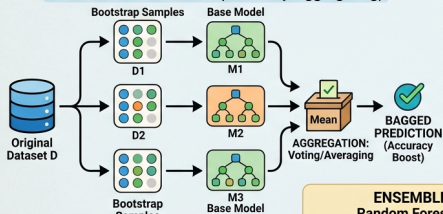
Ensemble methods combine predictions from multiple models to make a more accurate and robust final prediction.

UNDERSTANDING THE ENSEMBLE METHOD: COMBINING WEAK PREDICTORS FOR STRONGER RESULTS

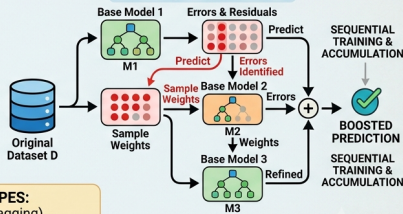
CONCEPT: UNIFYING WEAK PREDICTORS



TECHNIQUE 1: BAGGING (Bootstrap Aggregating)



TECHNIQUE 2: BOOSTING

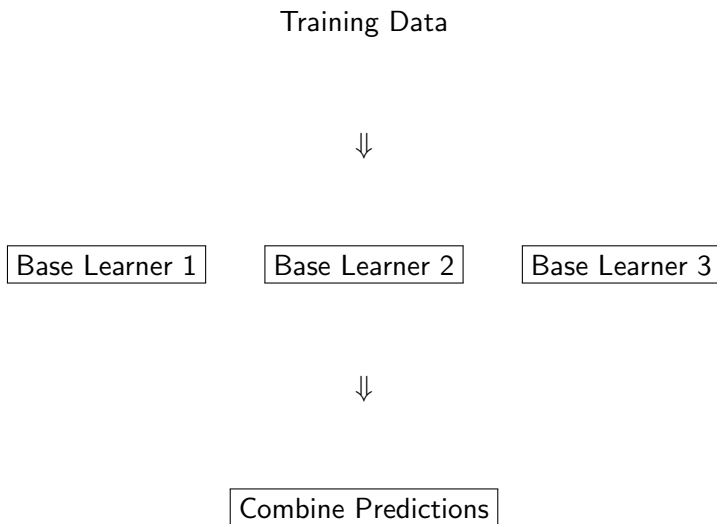


ENSEMBLE TYPES:
Random Forest (Bagging),
Gradient Boosting (AdaBoost, XGBoost)

LOW ACCURACY
(Single Model)

REDUCED VARIANCE (Bagging) → REDUCED BIAS (Boosting) → HIGH ACCURACY
(Ensemble Method)

How Ensemble Learning Works



Types of Ensemble Learning

Ensemble learning combines multiple machine learning models to produce a more accurate and robust prediction than a single model.

The four major types of ensemble learning are:

① **Bagging (Bootstrap Aggregating)**

- Trains multiple models independently using bootstrap samples.
- Reduces variance and improves stability.

② **Boosting**

- Trains models sequentially, where each model corrects the mistakes of the previous one.
- Reduces bias and improves prediction accuracy.

③ **Voting**

- Combines predictions from multiple independently trained models.
- Uses majority voting or probability averaging.

④ **Stacking**

- Uses a meta-learner to combine predictions from multiple base models.
- Learns the optimal way to combine different models.

Bagging (Bootstrap Aggregating)

Idea: Train multiple models independently using different bootstrap samples.

How it works

- 1 Generate multiple bootstrap samples from the original dataset.
- 2 Train one model on each bootstrap sample.
- 3 Each model makes its own prediction.
- 4 Combine predictions using:
 - Majority voting (Classification)
 - Averaging (Regression)

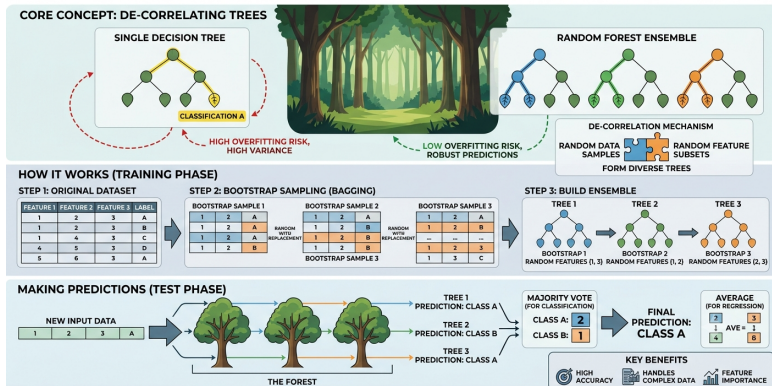
Advantages

- Reduces variance.
- Prevents overfitting.
- Models can be trained in parallel.

Example: Random Forest

Random Forest

UNDERSTANDING RANDOM FOREST: AN ENSEMBLE LEARNING ALGORITHM



Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification and regression performance. The final prediction is obtained by:

- Majority voting (Classification)
- Averaging predictions (Regression)

Boosting

Idea: Train models sequentially, where each new model focuses on correcting the mistakes made by previous models.

How it works

- 1 Train the first model.
- 2 Identify incorrectly predicted samples.
- 3 Increase the importance (weight) of those samples.
- 4 Train the next model using the updated weights.
- 5 Repeat until all models are trained.

Advantages

- Reduces bias.
- Improves prediction accuracy.
- Learns difficult patterns effectively.

Examples: AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost

AdaBoost (Adaptive Boosting)

Main Idea

- AdaBoost combines multiple weak learners to create a strong classifier.
- Initially, all training samples have equal weights.
- After each iteration, the weights of **misclassified samples are increased**.
- The next weak learner focuses more on these difficult samples.
- Final prediction is obtained using a weighted vote of all learners.

Advantages

- Simple and effective.
- Improves weak learners.
- Works well on small and clean datasets.

AdaBoost Algorithm

Input: Training set $\{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \{-1, +1\}$

① Initialize sample weights:

$$w_i = \frac{1}{N}, \quad i = 1, \dots, N$$

② For $t = 1, \dots, T$:

- Train a weak learner $h_t(x)$ using the weighted data.
- Compute the weighted classification error.
- Compute the learner weight.
- Update sample weights.
- Normalize the weights.

③ Combine all weak learners using weighted voting.

AdaBoost: Mathematical Formulation

Step 1: Weighted Error

$$\epsilon_t = \sum_{i=1}^N w_i I(y_i \neq h_t(x_i))$$

where $I(\cdot)$ is the indicator function.

Step 2: Compute Learner Weight

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

A classifier with lower error receives a larger weight.

Step 3: Update Sample Weights

$$w_i \leftarrow w_i \exp(-\alpha_t y_i h_t(x_i))$$

Normalize:

$$w_i = \frac{w_i}{\sum_{j=1}^N w_j}$$

Misclassified samples receive higher weights, while correctly classified samples receive lower weights.

AdaBoost: Final Classifier

The final prediction is obtained by combining all weak learners.

$$H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right)$$

where

- $h_t(x)$: prediction of the t^{th} weak learner
- α_t : importance (weight) of the learner
- T : number of weak learners

Key Idea

AdaBoost sequentially trains weak learners. Each new learner focuses more on previously misclassified samples. The final decision is a weighted vote, where more accurate learners contribute more.

Main Idea

- Models are trained sequentially.
- Each new model learns from the **residual errors** of the previous model.
- The prediction is gradually improved by reducing these errors.
- Uses gradient descent to minimize the loss function.

Advantages

- High prediction accuracy.
- Flexible for classification and regression.
- Can optimize different loss functions.

XGBoost (Extreme Gradient Boosting)

Main Idea

- An optimized implementation of Gradient Boosting.
- Includes regularization to reduce overfitting.
- Supports parallel computation and tree pruning.
- Automatically handles missing values.

Advantages

- Faster training.
- Excellent prediction accuracy.
- Widely used in data science competitions.

LightGBM (Light Gradient Boosting Machine)

Main Idea

- Developed by Microsoft.
- Uses histogram-based learning for faster training.
- Grows trees **leaf-wise** instead of level-wise.
- Designed for large-scale datasets.

Advantages

- Very fast and memory efficient.
- High accuracy on large datasets.
- Suitable for millions of training samples.

CatBoost (Categorical Boosting)

Main Idea

- Developed by Yandex.
- Designed to work directly with categorical features.
- Uses ordered boosting to reduce prediction bias.
- Requires minimal data preprocessing.

Advantages

- Automatically handles categorical variables.
- Reduces overfitting.
- High accuracy with mixed numerical and categorical data.

Comparison of Boosting Algorithms

Algorithm	Main Idea	Speed	Best For
AdaBoost	Reweight misclassified samples	Medium	Small datasets
Gradient Boosting	Learn residual errors	Slow	General prediction
XGBoost	Optimized Gradient Boosting	Fast	High accuracy
LightGBM	Leaf-wise tree growth	Very Fast	Large datasets
CatBoost	Handles categorical features	Fast	Categorical data

Voting Ensemble

Idea: Combine predictions from multiple independently trained models to improve prediction accuracy.

Workflow

- 1 Train multiple models (same or different algorithms).
- 2 Each model predicts the output.
- 3 Combine predictions:
 - **Hard Voting:** Majority vote.
 - **Soft Voting:** Average predicted probabilities.

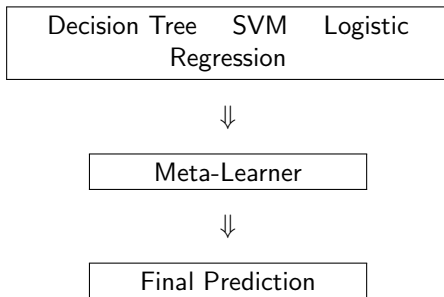
Example

Classifier	Prediction
Decision Tree	Yes
SVM	Yes
KNN	No
Logistic Regression	Yes
Final Prediction	Yes

Stacking: Working Process

Steps

- 1 Train multiple base models.
- 2 Each model makes predictions.
- 3 Use predictions as new features.
- 4 Train a meta-learner.
- 5 Generate the final prediction.



Comparison of Ensemble Methods

Method	Training	Main Idea	Example
Bagging	Parallel	Reduce variance	Random Forest
Boosting	Sequential	Reduce bias by learning from mistakes	AdaBoost, XGBoost
Voting	Parallel	Combine predictions	Voting Classifier
Stacking	Parallel	Learn how to combine models	Meta-Learner

Key Takeaways

- Ensemble methods combine multiple models to improve performance.
- Bagging trains models independently on bootstrap samples to reduce variance.
- Boosting trains models sequentially, focusing on previous mistakes to reduce bias.
- Voting combines predictions using majority voting or probability averaging.
- Stacking uses a meta-learner to learn the best combination of multiple models.